

Employee Salaries Project Documentation

Main Objective

The main objective of this project is to investigate the impact of different employee-related parameters on the salary received by each individual.

The analyzed parameters are: id, gender, bdate, educ, jobcat, salary, salbegin, jobtime, prevexp, minority, employee_pro.

Data Cleaning

The dataset was already relatively clean. Only one row contained a null value and was removed from the analysis.

Feature Engineering

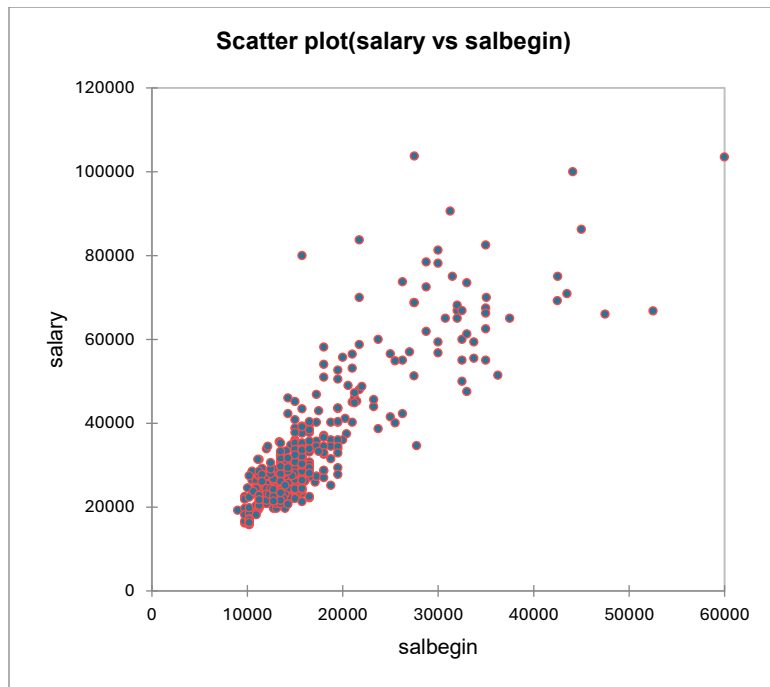
To extract more information from the dataset, the following derived features were created from the original variables:

- Age
- exp_ratio
- total_exp
- salary_growth

Salary and Salbegin:

$r = 0.861$, $p < 0.0001$

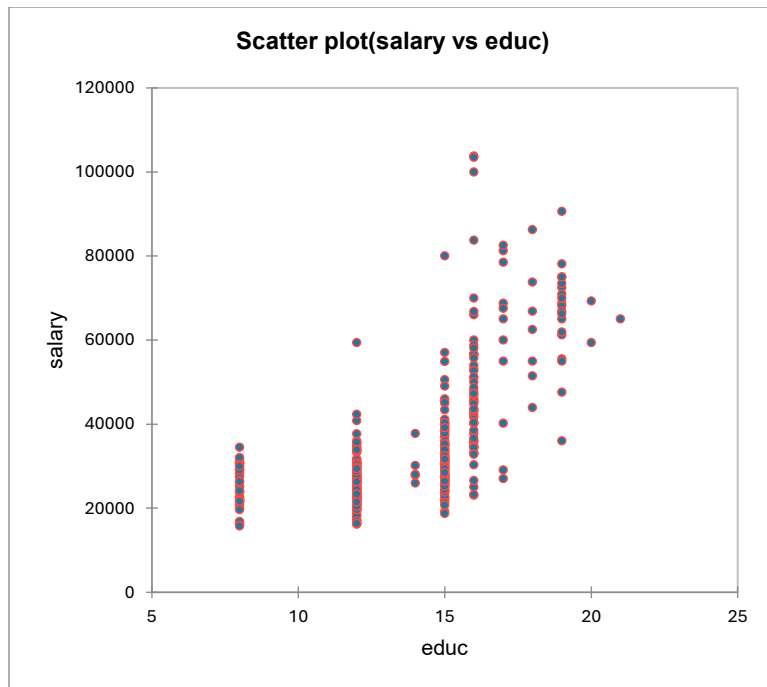
A **very strong positive correlation** was found between current salary and starting salary, implying that employees who started with higher salaries tend to maintain higher salary levels over time.

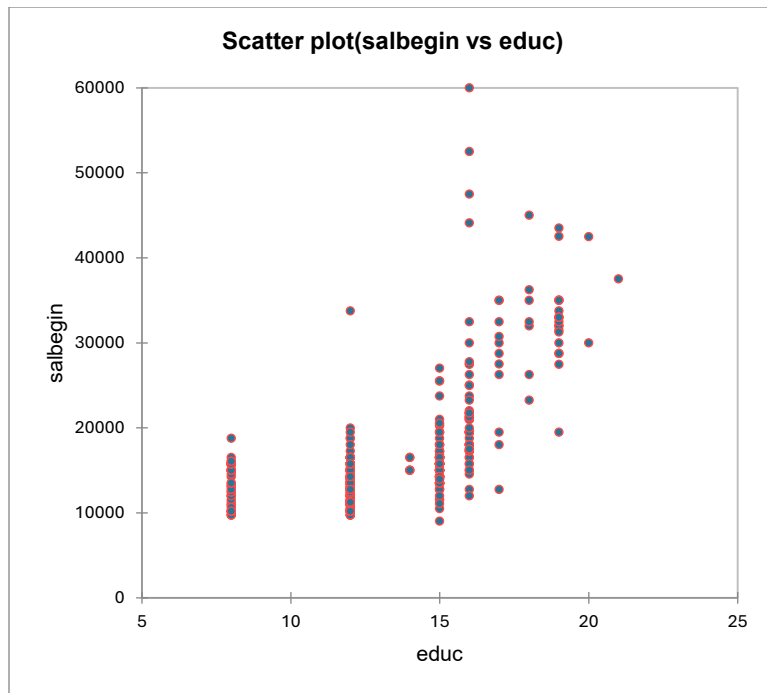


Salary and Education:

$r = 0.642, p < .001$

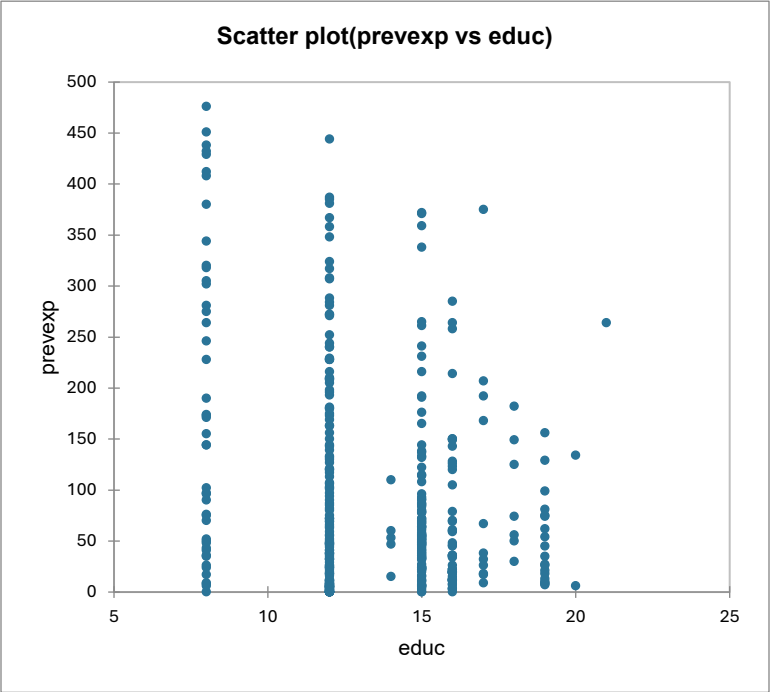
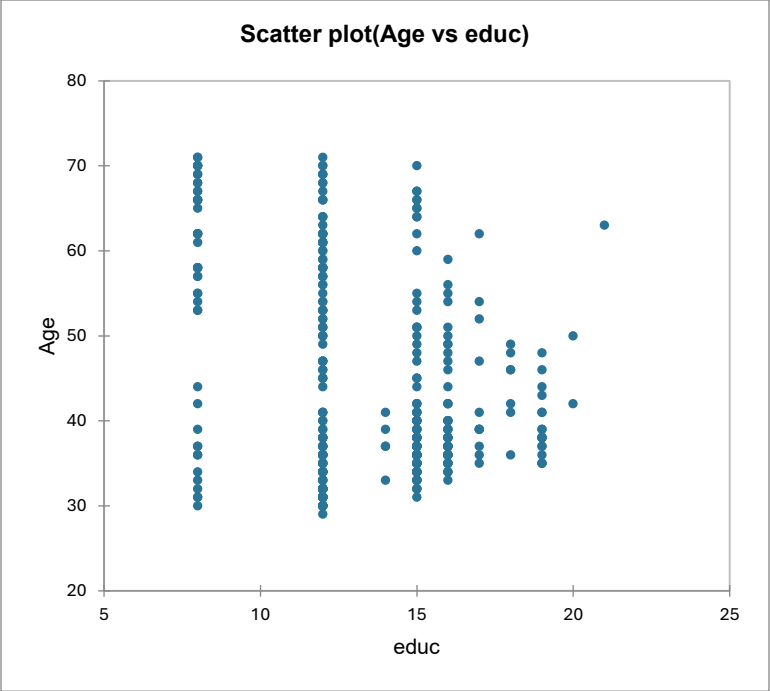
The results revealed a **significant positive correlation between education and current salary**, indicating that employees with higher educational attainment tend to receive higher current salaries.





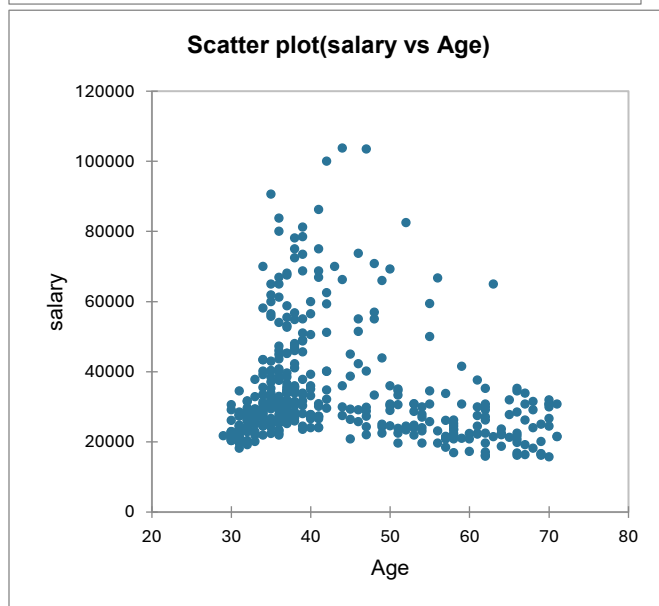
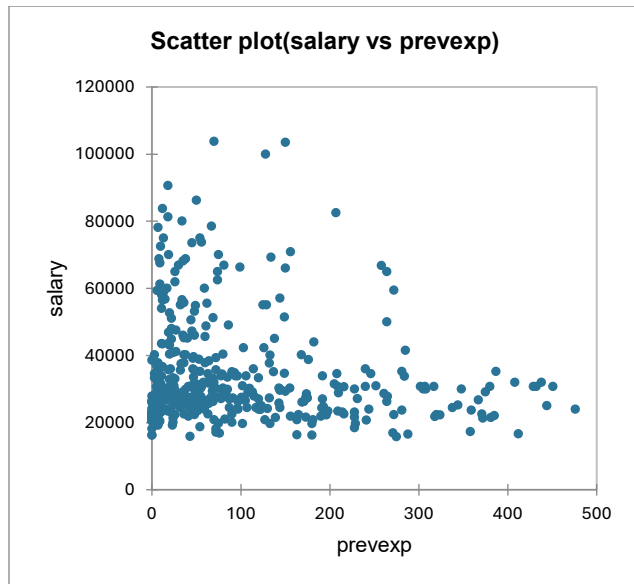
Education, prevexp and age:

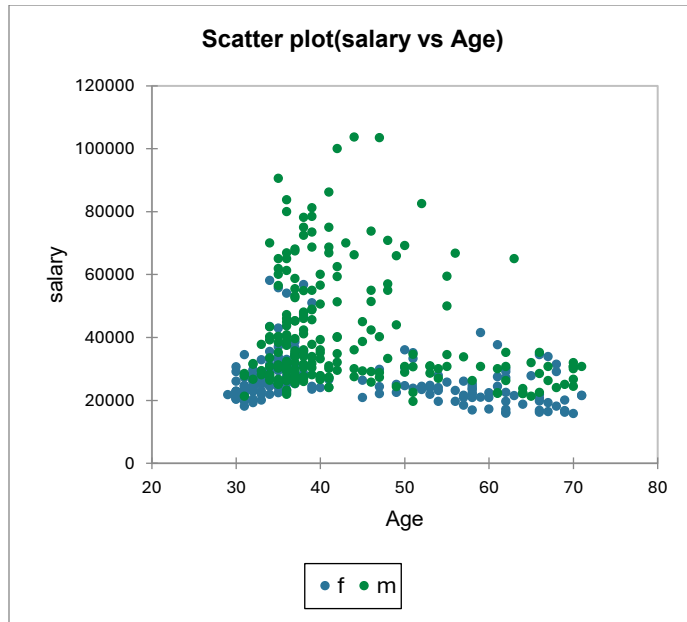
Education showed **weak negative correlations with previous work experience** ($r = -0.271$, $p < .001$) and **age** ($r = -0.290$, $p < .001$). These findings indicate that employees with higher educational levels tend to be younger and have less prior work experience.



Salary, Prevexp and age:

Current salary was **negatively weak correlated** with previous work experience ($r = -0.127$, $p = .010$) and age ($r = -0.168$, $p = .001$). Although these relationships were statistically significant, their magnitudes were relatively weak.



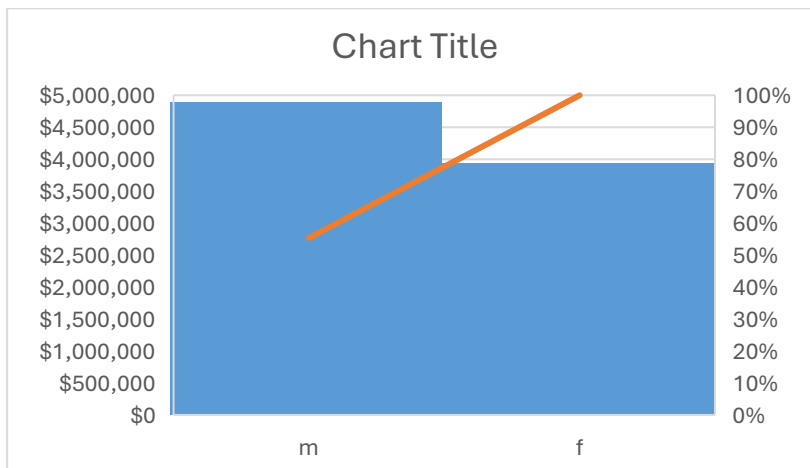
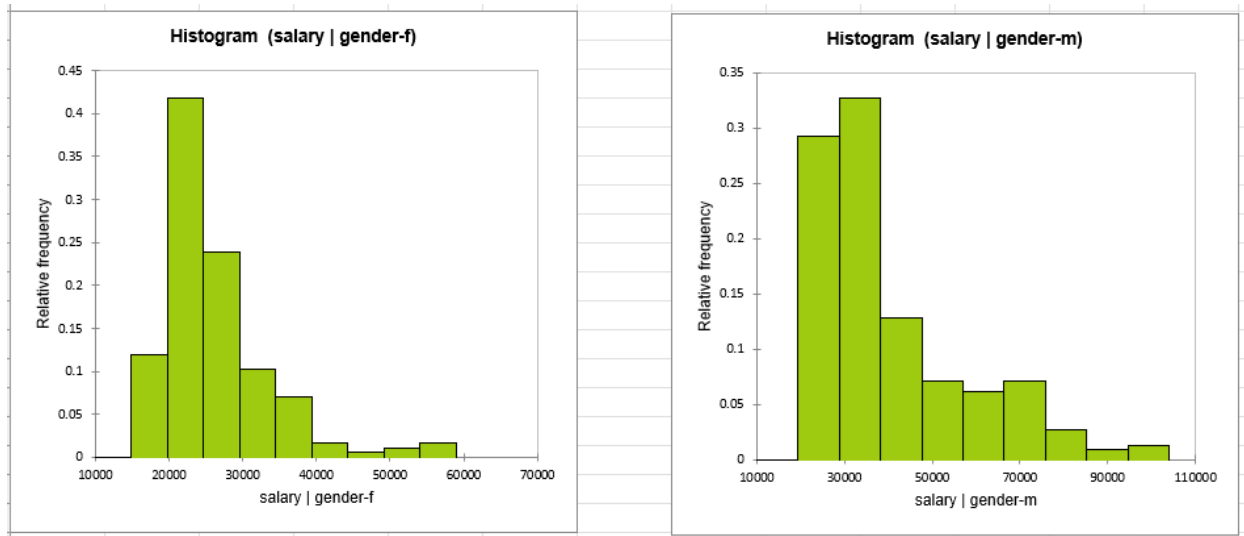


Salary and Gender:

The results showed a **significant negative biserial correlation between gender and current salary** ($r = -0.445$, $p < .001$).

Since the p-value was lower than the significance level ($\alpha = 0.05$), the null hypothesis (H_0 : **salary and gender are not associated**) was rejected, and the alternative hypothesis (H_a : **salary and gender are associated**) was accepted.

Specifically, the results suggest that there is a statistically significant difference in current salary between male and female employees. The magnitude of the correlation indicates a **moderate relationship** between gender and current salary.

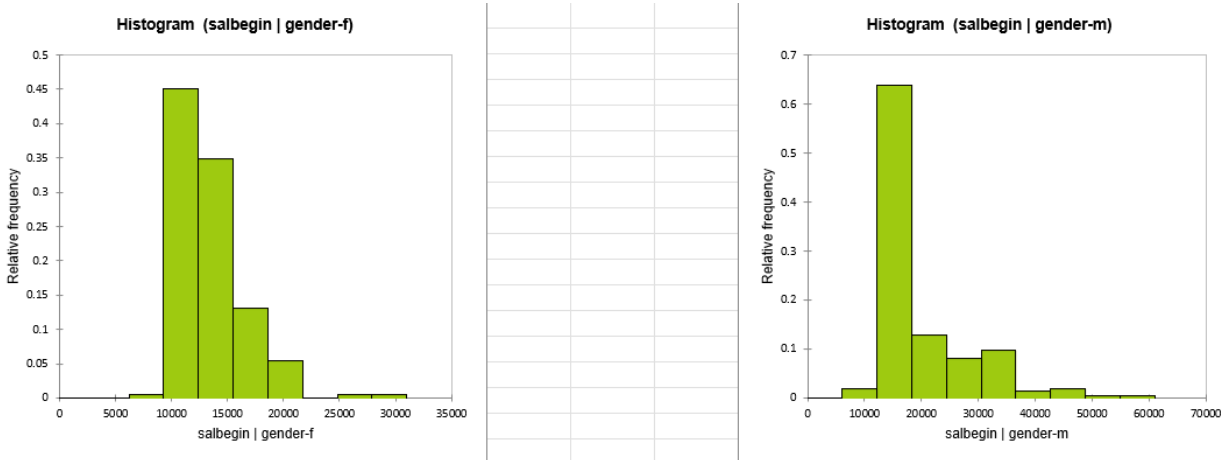


Salbegin and Gender:

A biserial correlation analysis revealed a **significant negative biserial correlation between gender and starting salary** ($r = -0.460, p < .001$).

Because the p-value was below 0.05, the null hypothesis (**H_0 : starting salary and gender are not associated**) was rejected. Therefore, gender and starting salary are significantly related.

The negative correlation suggests that starting salary differs significantly between male and female employees. The strength of the correlation indicates a **moderate association**, meaning that gender explains a meaningful proportion of the variation in initial salary levels.



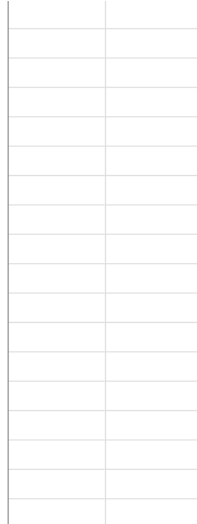
Salary and Minority:

The results showed a **significant positive biserial correlation between minority status and current salary** ($r = 0.150$, $p = .004$).

Since the p-value was lower than the significance level ($\alpha = 0.05$), the null hypothesis (H_0 : **salary and minority status are not associated**) was rejected, and the alternative hypothesis (H_a : **salary and minority status are associated**) was accepted.

The positive correlation indicates that minority status is statistically associated with differences in current salary. Because minority status was coded as 0 (non-minority) and 1 (minority), the positive coefficient suggests that employees classified as minorities tend to have higher salaries compared with non-minority employees in this sample.

However, the correlation coefficient is relatively small ($r = 0.150$), indicating a **weak relationship** between minority status and salary. Although the association is statistically significant, its practical effect appears limited.



Regression of variable salary:								
Goodness of fit statistics (salary):								
Observations	410	Analysis of variance (salary):						
Sum of weights	410							
DF	408	Source	DF	Sum of squares	Mean squares	F	Pr > F	
R ²	0.198	Model	1	19807701442.778	19807701442.778	100.925	<0.0001	
Adjusted R ²	0.196	Error	408	80074818468.502	196261809.972			
MSE	196261809.972	Corrected Total	409	99882519911.281				
RMSE	14009.347	Computed against model $Y = \text{Mean}(Y)$						
MAPE	28.655							
DW	1.900							
Cp	2.000							
AIC	7830.929							
SBC	7838.961							
PC	0.810							

Are the average managers' salaries for women and men different?

The results of the **ANOVA test** in the sample of 71 participants indicate that the mean salaries of female and male managers differ significantly ($F = 11.02$, $p = 0.001$). The average salary of men is approximately 63,546, while that of women is approximately 46,418, meaning that women receive, on average, about 17,129 units less. Gender explains approximately 13.8% of the variation in salaries. Given that the assumption of normality of the residuals was violated, it would be advisable to confirm this finding using a nonparametric test as well.

Analysis of variance (salary):						
Source	DF	Sum of squares	Mean squares	F	Pr > F	
Model	1	2305787284.568	2305787284.568	11.020	0.001	
Error	69	14436935814.023	209230953.826			
Corrected	70	16742723098.592				
Computed against model $Y = \text{Mean}(Y)$						
Model parameters (salary):						
Source	Value	Standard error	t	Pr > t	Lower bound (95%)	Upper bound (95%)
Intercept	63546.371	1837.034	34.592	<0.0001	59881.589	67211.153
gender-f	-17128.593	5159.707	-3.320	0.001	-27421.927	-6835.259
gender-m	0.000	0.000				

The **Mann-Whitney U test** comparing the salaries of male and female managers indicated that there is a statistically significant difference between the two groups ($U = 471$, $Z = 3.31$, $p = 0.001$). Therefore, the assumption of equal salaries between the two groups is rejected. The mean salary of male managers (63,546) was significantly higher than that of female managers (46,418), indicating the presence of a significant gender pay gap in this sample.

The sample sizes of the two groups are not equal:

- Men: 62 participants
- Women: only 9 participants

This imbalance may affect the statistical power of the test. However, because the p-value is very small (0.001), the observed difference is large enough to remain statistically significant even with the smaller female group.

Summary statistics:							
Variable	Observations	Obs. with missing	Obs. without	Minimum	Maximum	Mean	Std. deviation
salary_m	62	0	62	#####	#####	#####	#####
salary_f	9	0	9	#####	58125.000	#####	8612.897
Mann-Whitney test / Two-tailed test:							
U	471						
U (standardized)	3.310						
Expected U	279.000						
Variance (U)	3347.326						
p-value (Two-tailed)	0.001						
alpha	0.050						
The exact p-value could not be computed. An approximation has been used to compute the p-value.							
Test interpretation:							
H0: The difference of location between the samples is equal to 0.							
Ha: The difference of location between the samples is different from 0.							
As the computed p-value is lower than the significance level alpha=0.05, one should reject the null hypothesis H0, and accept the alternative hypothesis Ha.							

Is the salary growth equal for those with less than 16 years of education (up to a bachelor's degree) and those with more than that?

To examine that, an independent samples **t-test** was initially considered. However, because the data violated the assumption of normality, the non-parametric Mann–Whitney U test was used as an alternative.

Growth_Grater1: Growth_Less16

Sample 1: Workbook = employee_pro.xlsm / Sheet = Salary growth & Education / Range = 'Salary growth' / Sample 2: Workbook = employee_pro.xlsm / Sheet = Salary growth & Education / Range = 'Salary growth'

Hypothesized difference (D): 0

Significance level (%): 5

Population variances for the t-test:

Summary statistics:

Variable	Observations	Obs. with missing data	Obs. without missing data	Minimum	Maximum	Mean	Std. deviation
Growth_Grater16	93	0	93	6870.000	7570.000	7220.000	2709.033
Growth_Less16	317	0	317	5550.000	6250.000	5900.000	2709.033

Normality test:

Shapiro-Wilk test (Growth_Grater16):

W	0.950
p-value	0.001
alpha	0.050

Test interpretation:

H0: The residuals follow a Normal distribution.

Ha: The residuals do not follow a Normal distribution.

As the computed p-value is lower than the significance level alpha=0.05, one should reject the null hypothesis H0, and accept the alternative hypothesis Ha.

Summary statistics

Variable	Observations	Obs. with missing data	Obs. without missing data	Minimum	Maximum	Mean	Std. deviation
Growth_C	93	0	93	6870.000	#####	#####	#####
Growth_L	317	0	317	5550.000	#####	#####	5709.033

Normality test:

Shapiro-Wilk test (Growth_Grater16):

W	0.950
p-value (	0.001
alpha	0.050

Test interpretation:

H0: The residuals follow a Normal distribution.

Ha: The residuals do not follow a Normal distribution.

As the computed p-value is lower than the significance level $\alpha=0.05$, one should reject the null hypothesis H_0 , and accept the alternative hypothesis H_a .

The results of the Mann–Whitney U test revealed a statistically significant difference between the two groups ($U = 26,080.5$, $Z = 11.285$, $p < 0.0001$). Therefore, the null hypothesis was rejected, indicating that the distributions of salary growth for employees with less than 16 years of education and those with more than 16 years of education are significantly different.

Summary statistics:							
Variable	Observations	Obs. with missing data	Obs. without missing data	Minimum	Maximum	Mean	Std. deviation
Growth_Grater16	93	0	93	6870.000	76240.000	28627.903	12922.503
Growth_Less16	317	0	317	5550.000	64250.000	13649.637	5709.033
Mann-Whitney test / Two-tailed test:							
U	26080.500						
U (standardized)	11.285						
Expected value	14740.500						
Variance (U)	1009620.524						
p-value (Two-tailed)	<0.0001						
alpha	0.050						
The exact p-value could not be computed. An approximation has been used to compute the p-value.							
Test interpretation:							
H0: The difference of location between the samples is equal to 0.							
Ha: The difference of location between the samples is different from 0.							
As the computed p-value is lower than the significance level $\alpha=0.05$, one should reject the null hypothesis H0, and accept the alternative hypothesis Ha.							

Ancova with Age VS without Age:

Age was removed from the ANCOVA model because it did not have a statistically significant effect on salary after controlling for the other predictors. In the first model, the coefficient for Age was **-34.25** with a **p-value of 0.513**, indicating that age did not contribute meaningfully to explaining salary variation. In addition, Age showed a very strong correlation with previous work experience (**prevexp**, $r = 0.813$), suggesting multicollinearity and substantial overlap in the information provided by these two variables. After removing Age, the model's explanatory power remained unchanged ($R^2 = \mathbf{0.812}$), while several fit statistics improved slightly (AIC decreased from **7247.68** to **7246.12** and RMSE decreased from **6829.31** to **6824.46**). Furthermore, the effect of previous experience became stronger and more statistically significant. Therefore, Age was excluded because it was a non-significant predictor that introduced redundancy, and its removal resulted in a more parsimonious model without any loss of predictive performance.